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# Spatial land suitability modeling for livestock production for closing yield gaps in Western parts of Ethiopia

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## ABSTRACT

Assessing potential land suitability is essential for effective land use planning aimed at maximizing livestock production while minimizing environmental impacts. The objective of this study was to assess the suitability of land for livestock production through the application of geospatial technologies in the western region of Ethiopia. The study integrated the Analytical Hierarchy Process (AHP) model with GIS techniques to generate land suitability classes for livestock production. Eight key parameters were considered: slope, land surface temperature (LST), agro-ecological zone (AEZ), rainfall, land use/land cover (LULC), distance to streams, distance to roads, and distance to markets. Land suitability for livestock production in the study area was classified into four categories: highly suitable, moderately suitable, marginally suitable, and unsuitable. The results revealed that about 66.1 km<sup>2</sup> (1.6%), 25.8 km<sup>2</sup> (0.6%), 315.3 km<sup>2</sup> (7.8%), and 20.1 km<sup>2</sup> (0.5%) were highly suitable for cattle, sheep, goats and camels production, respectively. However, about 48.6 km<sup>2</sup> (1.2%), 92 km<sup>2</sup> (2.3%), 6.4 km<sup>2</sup> (0.2%), and 65 km<sup>2</sup> (1.6%) were not suitable for cattle, sheep, goat and camel, respectively. Therefore, based on the study's findings, it is recommended that farmers and policymakers manage and utilize the identified suitable areas to enhance both the quantity and quality of livestock production in the region.

## IMPACT STATEMENT

Land suitability analysis for livestock production is crucial to enhance food security in the community. This study presents the spatial land suitability modeling for livestock production for closing yield gaps in western parts of Ethiopia. Geographic information system and remote sensing application with analytical hierarchy process (AHP) were used in this study. Eight factors such as slope, land surface temperature (LST), agro-ecological zone (AEZ), rainfall, land use land cover (LULC), distance to streams, distance to roads, and distance to markets were used. The results shows that about 1.6%, 0.6%, 7.8%, and 0.5% were highly suitable for cattle, sheep, goats, and camel production, respectively. The finding of this study is useful for farmers and decision-makers can manage and utilize the identified suitable areas to produce sufficient and high quality livestock production in the study area.

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## Introduction

Livestock production remains one of the world's most important sectors, contributing significantly to the economy, food security, and livelihoods. Globally, it provides employment for over 1.3 billion people and accounts for approximately 40% of total agricultural GDP (FAO, 1993). Moreover, livestock has been, is,

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and will continue to be central to the development of food systems, sustainability efforts, and global health security (Thornton, 2010). Globally, livestock accounts for approximately 40% of agricultural GDP and accounts for approximately 30% of agricultural GDP in developing countries. Livestock production is undertaken in a multitude of ways across the world, providing a large variety of goods and services in different sets of resources in a wide spectrum of agro-ecological and socio-economic conditions (Yitbarek & Berhane, 2014). Livestock are valuable assets in Africa, contributing to economic growth while improving the nutritional status of their owners.

Ethiopia is the first in Africa and tenth in the world with livestock population (Duguma et al., 2012; Erkyihun et al., 2022). The main reason for the country's livestock richness is the existence of various demographic conditions and agro ecological zonation, which creates the probability for the occurrence different species of livestock (Tegegne et al., 2010; Thornton et al., 2009). Based on this point of views different scholars' reports the country's livestock census from time to time. The previous findings revealed that the presence of 43.1 million heads of cattle, 23.6 million sheep, 18.6 million goats, 4.5 million donkeys, 1.7 million horses, 34.2 million chickens, and 4.9 million beehives (Aime & Yves, 2017).

Livestock is an integral part of the agriculture sector and contributes to 16.5% of the country's GDP, 47.7% of the agricultural GDP, 15% of export earnings and 30% of agricultural employment (Duguma, 2022; Kidanemariam & Fesseha, 2020). It also serves as sources of food, manure, raw materials, investment, cash income, productive traits, foreign exchange earnings, and social and cultural identity (Wudneh, 2022). The production and productivity of livestock farming were basically based up on the potential biological productivity and suitability of the land (Bogale & Erena, 2022). However, potentiality of the specific lands for livestock production was determined by various environmental parameters like land uses, and land cover (LULC) types, elevation, temperature, slope, soil types, rain fall, water accessibilities and other infrastructures such as markets, road and veterinary site availabilities.

Although several studies have assessed land suitability for livestock production in various regions of the country (Balew et al., 2022; Terfa & Suryabhadgavan, 2015), most have analyzed relevant factors independently, resulting in fragmented and less comprehensive evaluations. There is a clear need for an integrated analysis that combines multiple factors to provide a more holistic and accurate assessment of land suitability, as livestock production is influenced by the complex interplay of biophysical, climatic, and infrastructural variables. Furthermore, previous studies have largely overlooked critical parameters such as land surface temperature, which directly affects animal health, productivity, and habitat suitability. Incorporating land surface temperature alongside other key variables is essential for capturing the thermal environment's impact on livestock and for improving model precision. This study aims to address these gaps by employing a multi-criteria evaluation approach that integrates land surface temperature with other significant factors, thereby enhancing the reliability and applicability of land suitability assessments for sustainable livestock production planning.

Even though the country has high potential for livestock resources, the contribution from the sector to the national economy has been relatively modest compared to its production potential. An evaluation of potential land suitability is a core point for enhancements of livestock productivity status in a specific area by matching livestock species specific preferences. Improved livestock production was related with sustainable livelihood standard and developments in local community.

The second-most prominent system in Ethiopia is pastoral and agro-pastoral livestock production, which is concentrated mostly in the south and east of the country in Afar, Somali, Southern Oromia (Borana), Kereyou in East Shoa, and South Omo in Southern Nations, Nationalities, and Peoples' Region (SNNPR). However, the habit of livestock production was undeveloped due to a lack of sufficient information on the evaluation of potential land suitability for livestock production particularly in western parts of Ethiopia. In this study hybrid of Analytic Hierarchy Process (AHP) and Geographic information application were used for identifying optimal livestock suitability to close yield gaps in the semi-arid region of Ethiopia. The application of the AHP for land suitability analysis has become increasingly popular due to its ability to integrate various criteria and decision-making processes in a systematic and transparent way (Dos Santos et al., 2019). Land suitability for livestock, AHP provides a robust framework for evaluating multiple environmental, biological, and socio-economic factors that influence the optimal use of land (Zhang et al., 2023). Previous studies have demonstrated the effectiveness of AHP in integrating diverse data sets, such as soil quality, climate conditions, topography, and water availability, to

assess the most suitable land areas for livestock grazing, farming, and other related activities (Peng et al., 2024; Saaty, 1980; Yalew et al., 2016). These methodologies are particularly useful in our study area with limited data availability, as AHP allows for the inclusion of expert opinions and local knowledge, enhancing the accuracy and relevance of land suitability assessments for livestock production (Gallego et al., 2019).

In this study, Geographic Information Systems (GIS) were integrated with the Analytical Hierarchy Process (AHP) to spatially analyze land suitability for livestock production. GIS enables the visualization, analysis, and integration of multiple biophysical and infrastructural factors into a spatial model, helping to identify areas with the highest potential for sustainable livestock farming. Previous research by Gallego et al. (2019) demonstrated that the combined use of GIS and AHP can generate accurate land suitability maps by evaluating criteria such as forage availability, water access, and environmental sustainability. These integrated approaches enhance decision-making, especially in regions like southwestern Ethiopia, where traditional land suitability assessments are often hindered by limited and inconsistent data. A key strength of AHP is its adaptability to various livestock farming systems such as cattle, sheep, and goats making it applicable across diverse geographic and ecological settings. Despite its potential, limited research has been conducted on land suitability analysis for livestock production in Ethiopia. In this context, geospatial technology offers a valuable tool for mapping the spatial distribution and extent of suitable land, providing insights for the effective use of highly productive areas. Therefore, this study aims to address the existing research gap by modeling spatial land suitability for livestock production using geospatial techniques in the western part of Ethiopia, with the ultimate goal of contributing to improved livestock yields and sustainable resource utilization.

## Materials and methods

### Description of the study area

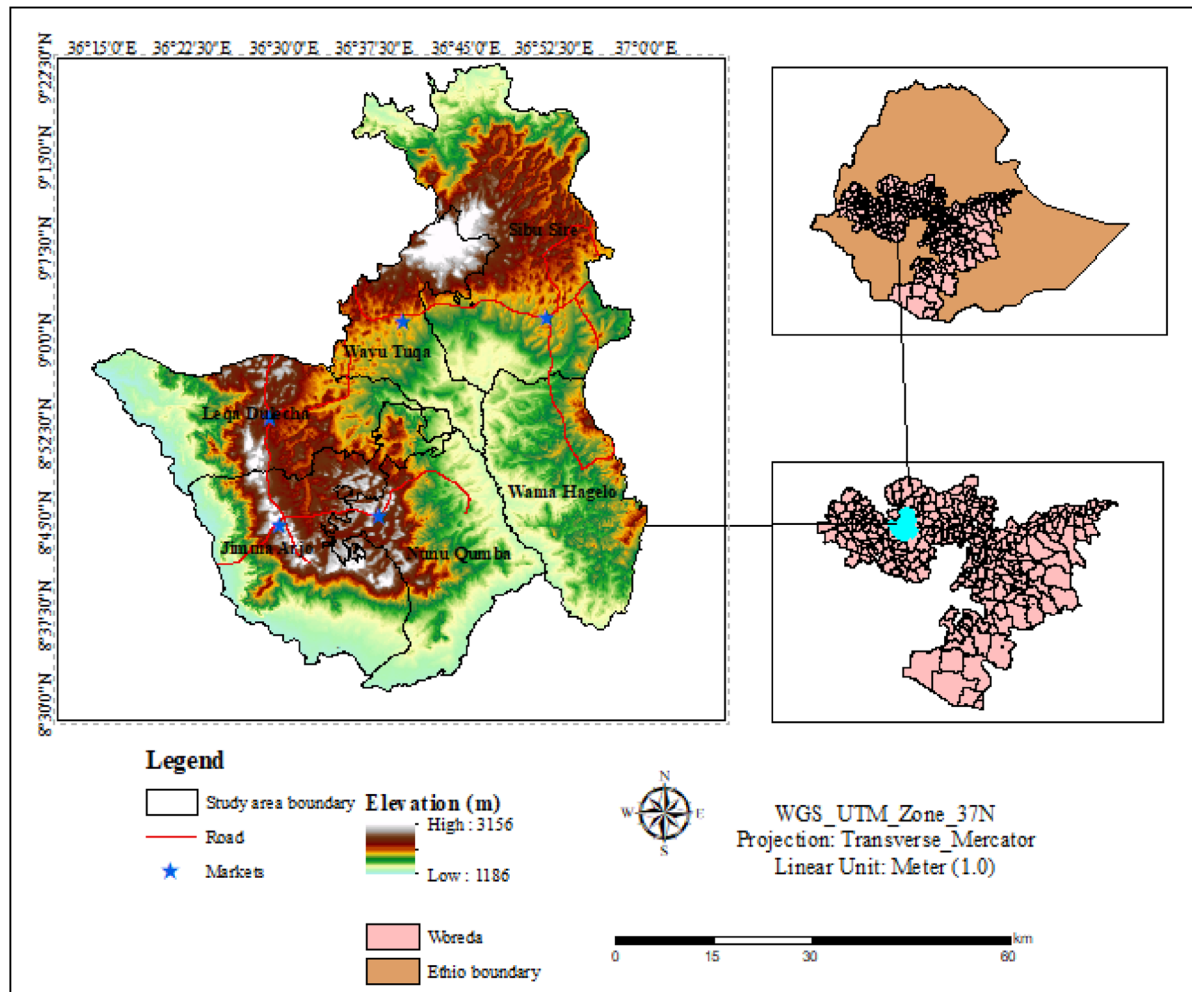
The study was conducted in the southwestern parts of the east Wollega zone in six selected districts (Sibu Sire, Wayu Xuqa, Jimma Arjo, Nunu Kumba, Leqa Dulecha, and Wama Hagelo). Selection of the representative districts based on agro ecological conditions for livestock production. Due to the fact that livestock production continues to be one of the most crucial sectors in terms of the global economy, food security, and human livelihoods.

These selected districts were existed in lowland (*Kola*), middle land (*Woina dega*) and Highland (*Dega*) by local name. Geographically, the study area was situated between 8°30'00" and 9°27'30"N and 36°15'00" and 37°0'00"E with total area of 4049.2km<sup>2</sup> (Figure 1). The topographic variation of the selected districts ranges from 1186 to 3156m above the mean sea level. The selected districts received maximum rainfall during the summer season (June to September) with 1121 to 1705mm. The maximum and minimum temperatures of the study area were 39°C and 15°C, respectively, with the mean temperature of 29°C.

### Data sources and description

In this study, the combination of topographical, physical, bio-physical, hydrological, and socio economic data were used to analyze the land suitability analysis for livestock production in the southwestern parts of Ethiopia. Accordingly, eight factors, slope, agro ecological zone (AEZ), Streams, LULC, land surface temperature (LST), rainfall, roads and markets were used to analysis the potential land suitability for livestock production. A Digital Elevation Model (DEM) for the study region was obtained from the ASTER database via their online platform <https://earthexplorer.usgs.gov/> that was used to generate slope, AEZ and streams parameters. Landsat image OLI (Operational Land Imager) and TIRS (Thermal Infrared Sensor) of 2022 was downloaded from the United States Geological Survey (USGS) website via <https://www.usgs.gov/> for classification of LULC and retrieval of LST for this study. Consequently, the rainfall of the study area was obtained from the national metrological agency (NMA) of Ethiopia at and near stations of the study area to calculate annual rainfall. In addition, infrastructure data (road and market) were obtained from EMA for the study area (Table 1 and Figure 2).





**Figure 1.** Study area boundary (left panel), Ethiopia boundary and Oromia woreda (right-top), and Oromia boundary (right-bottom).

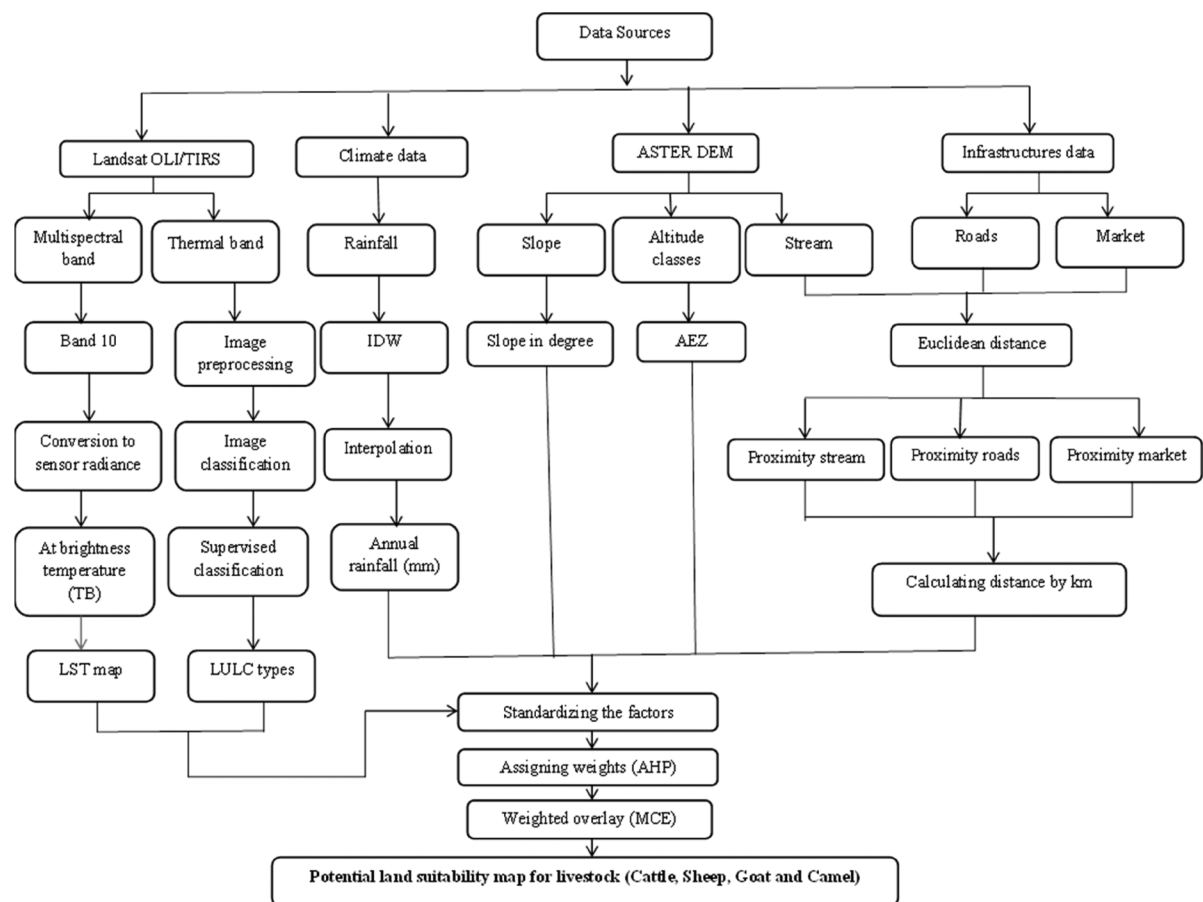
**Table 1.** Data types and sources.

| S/no | Data types            | Descriptions       | Sources                            |
|------|-----------------------|--------------------|------------------------------------|
| 1    | Climate               | Rainfall           | National metrological agency (NMA) |
| 2    | ASTER DEM             | Stream, AEZ, Slope | ASTER website                      |
| 3    | Infrastructure        | Road and market    | Ethiopian National mapping agency  |
| 4    | Landsat OLI/TIRS 2022 | LULC, LST          | Earth explorer (USGS website)      |

From the software packages, ArcGIS 10.3 for analyzing and visualizing of spatial data and ERDAS Imagine 2015 for LULC classification and accuracy assessment were used. In addition, IDRISI Selva 17 was used for the calculation of pairwise comparison and assigning weights while ArcSWAT 10.3 was used for the generation of the drainage (stream) networks. Finally, Google Earth Pro was used for LULC accuracy assessment and final result validation.

### Method of data analysis

Biophysical data (slope, AEZ, streams, and LULC), climate data (land surface temperature [LST] and rainfall), and infrastructure data (roads and markets) were analyzed within an ArcGIS environment to assess land suitability for livestock production (Sang et al., 2022). The rating ranges for all parameters are detailed in Table 2.



**Figure 2.** Methodological flowchart of study.

### Slope

Slope is the main parameter used to analyze the suitability of land for livestock farming. Slope-based land suitability may be classified into different classes based on the types of livestock. The classified interval for cattle farming ranges from 0° to 8°, 8° to 16°, 16° to 25° and greater than 25°. In addition, suitable classes for sheep also range from 0° to 16°, 16° to 30°, 30° to 40° and greater than 40°. Consequently, goat and camel farming were performed in the slope ranges of 0° to 16°, 16° to 35°, 35° to 50° and greater than 50° and 0° to 6°, 6° to 12°, 12° to 20° and greater than 20°, respectively (De Marinis et al., 2022). These slope classifications help guide optimal land use decisions to maximize livestock productivity while minimizing environmental degradation.

### Land surface temperature

Land surface temperature is expressed in degrees Celsius, reflects the thermal condition of a study area. Variations in LST within the study area influence the classification of land suitability for livestock farming, as temperature affects many critical factors for livestock production (Rojas-Downing et al., 2017). Suitable temperature classes for cattle and sheep range from 15°C to 23°C, 23°C to 28°C, 28°C to 31°C, and above 31°C. In contrast, appropriate temperature ranges for goat and camel rearing are 15°C to 25°C, 25°C to 28°C, 28°C to 31°C, and 31°C to 39°C (Yu et al., 2022).

### Agro ecological zone

Agro-ecologically, the study area was classified into three zones: highland, middle land, and low land. Variation in ecological zonation causes the existence of various livestock farms based on their

**Table 2.** Rating parameters for livestock production.

| Parameters/criteria |                           | Land suitability criteria rating for cattle, sheep, goat and camel |                          |                          |                      |
|---------------------|---------------------------|--------------------------------------------------------------------|--------------------------|--------------------------|----------------------|
|                     |                           | High suitable(S1)                                                  | Moderately suitable (S2) | Marginally suitable (S3) | Not suitable (SN)    |
| Cattle              | Slope (degree)            | 0–8                                                                | 8–16                     | 16–25                    | >25                  |
|                     | LST (°C)                  | 15–25                                                              | 25–28                    | 28–31                    | >31                  |
|                     | AEZ (meter and name)      | 1500–2300 (midland)                                                | 1186–1500 (lowland)      | 2300–3156 (highland)     | –                    |
|                     | Rainfall (mm)             | >1582                                                              | 1455–1582                | 1320–1455                | 1121–1320            |
|                     | LULC                      | Grassland                                                          | Wetland                  | Cultivated land          | FL, Set, BL          |
|                     | Proximity to Streams (km) | 0–0.5                                                              | 0.5–0.8                  | 0.8–1.5                  | >1.5                 |
|                     | Slope (degree)            | 0–16                                                               | 16–30                    | 30–40                    | >40                  |
| Sheep               | LST (°C)                  | 15–25                                                              | 25–28                    | 28–31                    | >31                  |
|                     | AEZ (meter and name)      | 2300–3156 (highland)                                               | 1500–2300 (midland)      | –                        | 1186–1500 (lowland)  |
|                     | Rainfall (mm)             | >1582                                                              | 1455–1582                | 1320–1455                | 1121–1320            |
|                     | LULC                      | Grassland                                                          | Wetland                  | Cultivated land          | FL, Set, BL          |
|                     | Proximity to streams (km) | 0–0.3                                                              | 0.3–0.5                  | 0.5–1                    | >1                   |
|                     | Slope (degree)            | 0–16                                                               | 16–35                    | 35–50                    | >50                  |
|                     | LST (°C)                  | 31–39                                                              | 28–31                    | 25–28                    | <25                  |
| Goat                | AEZ (meter and name)      | 1186–1500 (lowland)                                                | 1500–2300 (midland)      | –                        | 2300–3156 (highland) |
|                     | Rainfall (mm)             | 1121–1320                                                          | 1320–1455                | 1455–1582                | >1582                |
|                     | LULC                      | grassland                                                          | –                        | Cultivated land          | Set, FL, BL, WL      |
|                     | Proximity to streams (km) | 0–0.3                                                              | 0.3–0.5                  | 0.5–1                    | >1                   |
|                     | Slope (degree)            | 0–6                                                                | 6–12                     | 12–20                    | >20                  |
|                     | LST (°C)                  | 31–39                                                              | 28–31                    | 25–28                    | <25                  |
|                     | AEZ (meter and name)      | 1186–1500 (lowland)                                                | 1500–2300 (midland)      | –                        | 2300–3156 (highland) |
| Camel               | Rainfall (mm)             | 1121–1320                                                          | 1320–1455                | 1455–1582                | >1582                |
|                     | LULC                      | grassland                                                          | –                        | Cultivated land          | Set, FL, BL, WL      |
|                     | Proximity to Streams (km) | 0–10                                                               | 10–20                    | 20–25                    | >25                  |
| Common parameters   | Proximity to roads (km)   | 0–5                                                                | 5–10                     | 10–20                    | >20                  |
| Common parameters   | Proximity to markets (km) | 0–10                                                               | 10–20                    | 20–30                    | >30                  |

Set=settlement; FL=forest land; BL=bare land; WL=wetland; AEZ=Agro ecological zone; LST=land surface temperature; LULC=land use land cover.

appropriate ecological preferences. According to Getachew et al. (2022), middle land was highly suitable for cattle farming, while low land was highly suitable for camel farming. In addition, low land was not suitable for sheep farming, whereas high land was not suitable for goat and camel farming activities.

### Rainfall

Rainfall was included as a key parameter in this study due to its significant influence on livestock production under varying environmental conditions (Balew et al., 2022). Rainfall amounts within the study area were measured in millimeters and used to classify land suitability and determine appropriate livestock types. Rainfall data were spatially interpolated using the Inverse Distance Weighting (IDW) method based on records from eight meteorological stations located in and around the study area. The rainfall-based suitability classes for cattle and sheep farming range from greater than 1582mm, 1455 to 1582mm, 1320 to 1455mm, and 1121 to 1320mm. Conversely, suitability for goat and camel farming corresponds to lower rainfall ranges, effectively the inverse of those for cattle and sheep (Dronova et al., 2011; Du et al., 2024).

### Land use and land cover

Land use and land cover types such as settlements, bare land, wetlands, forest land, grassland, and cultivated lands play a crucial role in the sustainability of livestock farming in the study area. These LULC types vary in their suitability for different livestock species, influencing factors such as forage availability, shelter, and water access. According to Mosisa and Asefa (2020), grasslands are highly suitable for livestock farming due to their abundant forage resources and open space, which support

grazing and movement. In contrast, settlements and built-up areas are generally unsuitable because of limited space, human disturbance, and potential conflicts between livestock and residents. Additionally, wetlands and forest lands may offer seasonal grazing opportunities but can be limited by conservation restrictions and variability in forage quality. Cultivated lands can sometimes serve as supplementary grazing areas, especially post-harvest, but excessive crop production can reduce their availability for livestock. Understanding the spatial distribution and classification of these LULC types is essential for effective land management and the sustainable development of livestock production systems in the study area.

### ***Proximity to the stream***

Water availability and distance from the production site may limit the sustainability of livestock farming in the study area. However, distance-based land suitability may vary from one livestock species to another. The land suitability classes for cattle farming range from 0 to 0.5 km, 0.5 to 0.8 km, 0.8 to 1.5 km, and greater than 1.5 km, whereas those for sheep and goats range from 0 to 0.3 km, 0.3 to 0.5, 0.5 to 1 km, and greater than 1 km. In addition, distance-based suitable areas for camel rearing fall between the intervals of 1 to 10 km, 10 to 20 km, 20 to 25 km, and greater than 25 km from the stream (Pfeiffer et al., 2022).

### ***Distance from roads***

Distance from the main road was a critical factor in determining land suitability for various types of livestock farming. Accessibility to transportation infrastructure directly influences the ease of moving livestock to markets, accessing veterinary services, and obtaining feed and other essential inputs, thereby affecting the overall productivity and economic viability of livestock operations. In this study, land suitability classes based on proximity to main roads were categorized into four intervals: 0 to 5 km, 5 to 10 km, 10 to 20 km, and greater than 20 km (de Rivera et al., 2022). Areas located within 0 to 5 km of main roads are generally considered highly suitable due to better market access and service availability, whereas areas beyond 20 km are less favorable due to increased transportation costs and logistical challenges. This classification enables targeted planning and resource allocation to optimize livestock production while minimizing operational constraints related to infrastructure.

### ***Proximity to the market***

Distance from the market was another infrastructural parameter that could influence the acceptability of the study area for livestock farming. The area nearest to the market was more prioritized than a distant one as a suitable livestock farming site. According to Jabeen et al. (2020), market distance-based land suitability ranges from 0 to 10 km, 10 to 20 km, 20 to 30 km, and greater than 30 km.

### ***Multi criteria evaluation (MCE) for livestock land suitability***

Integration of GIS and AHP are powerful tools when combined for land suitability assessments. GIS (handles spatial data and mapping, while AHP helps in multi-criteria decision making by assigning weights to different factors. Their integration allows for a systematic and objective evaluation of land based on various criteria, leading to informed land-use planning and management decisions (Duc, 2006).

In the reclassification analysis, all eight factor maps, both vector and raster, were resampled to a 30-meter spatial resolution to facilitate more precise analysis and results. The AHP method of MCE analysis was applied by using the proposed 1 to 9 scientific ratio in order to calculate the criteria weight of the special layer to the evaluate potential area for livestock farming based on all selected ecological parameters (Saaty, 1990). The livestock land suitability potential was evaluated using the classification system outlined by the (Food and Agriculture Organization (FAO), 1993; Gao et al., 2024). Final land suitability for livestock production was classified into highly suitable, moderately suitable, marginally suitable and not suitable. The obtained calculated values for all targeted factors were compared with each other

**Table 3.** Pairwise comparison of the parameters.

| Factors  | Slope | LST  | AEZ  | Rainfall | LULC  | Stream | Road | Market | Weight |
|----------|-------|------|------|----------|-------|--------|------|--------|--------|
| Slope    | 1     | 2    | 2    | 2        | 2     | 3      | 3    | 4      | 0.24   |
| LST      | ½     | 1    | 2    | 2        | 2     | 3      | 3    | 4      | 0.2    |
| AEZ      | ½     | ½    | 1    | 2        | 2     | 2      | 3    | 3      | 0.15   |
| Rainfall | ½     | ½    | ½    | 1        | 2     | 2      | 3    | 3      | 0.13   |
| LULC     | ½     | ½    | ½    | ½        | 1     | 2      | 3    | 3      | 0.1    |
| Stream   | 1/3   | 1/3  | ½    | ½        | ½     | 1      | 2    | 3      | 0.08   |
| Road     | 1/3   | 1/3  | 1/3  | 1/3      | ½     | 1/2    | 1    | 2      | 0.06   |
| Market   | ¼     | ¼    | 1/3  | 1/3      | 1/3   | 1/3    | ½    | 1      | 0.04   |
| Σ=       | 3.91  | 5.41 | 7.16 | 8.66     | 10.66 | 13.83  | 18.5 | 23     | 1      |

LST=Land surface temperature; AEZ=Agro ecological zone; LULC=Land use land cover.

by using a pairwise comparison matrix to reclassify and weight them based on their relative importance and degree of influence (Kordi & Brandt, 2012; Moisa et al., 2022; Saaty, 1980). The eigenvectors of the factors were normalized to determine the relative relevance of each parameter. The pairwise comparison matrix method uses people and expert preferences (Morales & de Vries, 2021; Solomon & Lukas, 2024) for comparing the classes and making a matrix table (Table 3).

According to Moisa et al. (2022), the consistency index was calculated using the consistency ratio (CR) of 0.08, and the expected value must be less than 0.1 to be acceptable (Eqs. 1 and 2).

The consistency check formula is:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (2)$$

where  $\lambda_{\max}$  is the largest Eigen value of the pairwise comparison matrix and  $n$  is the number of classes.

According to Saaty (1980) consistency ratio (CR) calculated as follows:

$$CR = \frac{CI}{RI} \quad (2)$$

where RI is the ratio index/average value of CI for random matrices.

### Weighted overlay analysis

Based on their significance value from the aggregated criteria map, prioritization of all parameters was made in order to evaluate the study area's land suitability for livestock farming by using the land suitability index (SI) (Orhan, 2021).

Suitability Index was calculated as follows (Eq. 3):

$$SI = \sum Wi * Xi \quad (3)$$

where SI=Suitability Index,  $W_i$ =weight of factor  $i$ , and  $X_i$ =normalized criterion score of factors. Finally, the weight was assigned based on their degree of influence.

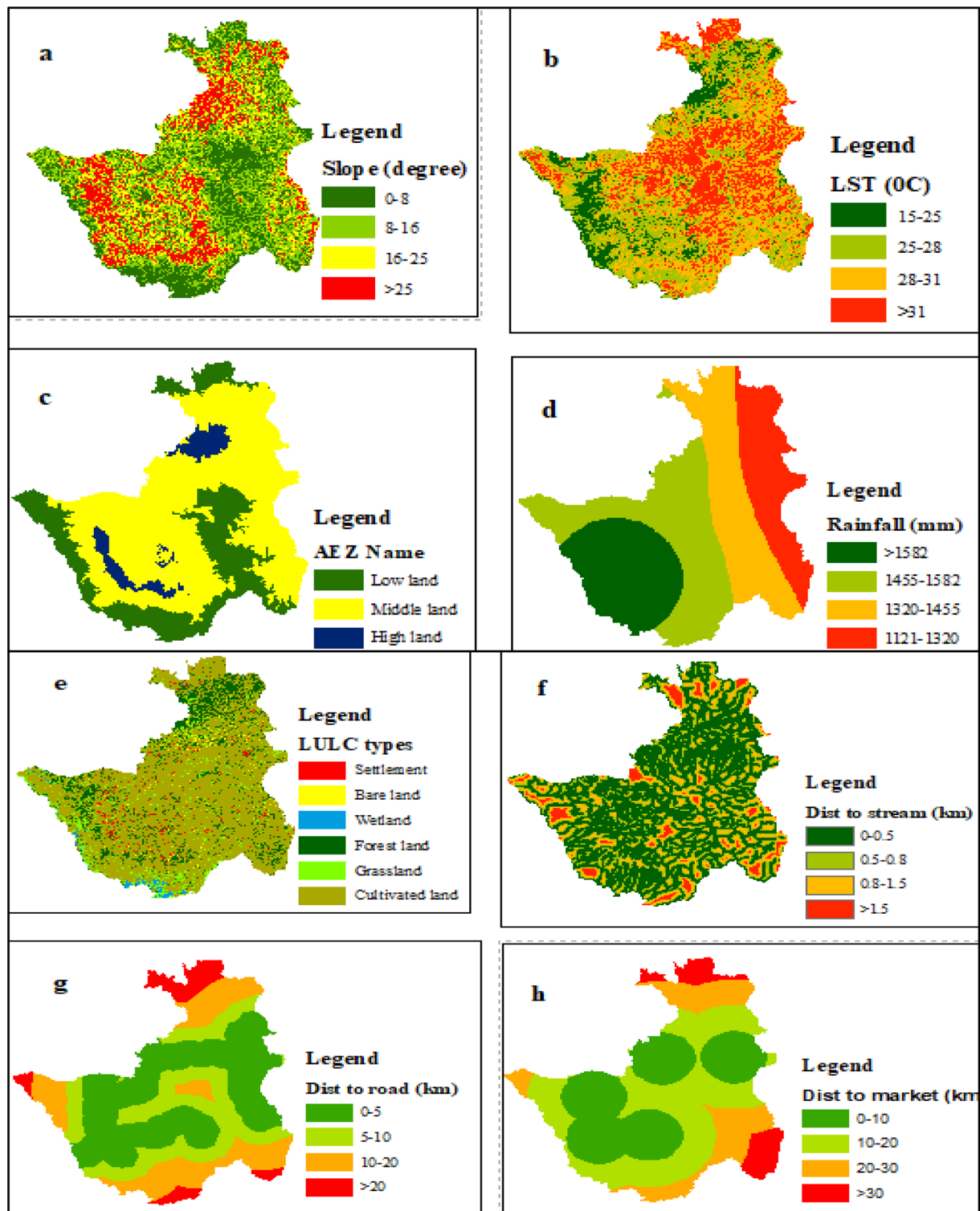
## Results and discussions

### Parameters used for land suitability for cattle production

#### Slope

Slope is a key factor in land suitability analysis for cattle production. Accordingly, slope was measured in degrees and classified into four categories: 0° to 8°, 8° to 16°, 16° to 25°, and greater than 25°. According to Duguma and Duguma (2022), gentle slopes (0°–8°) and medium slope (8°–16°) were highly suitable for cattle. However, slopes greater than 25° were not suitable for cattle production because gentle and medium slopes were more suitable than steep slopes. Consequently, the central and southern parts of





**Figure 3.** Parameters used for cattle farming land suitability.

the study area were highly suitable. However, the northern and western parts of the study area were not suitable for cattle production (Figure 3a).

#### *Land surface temperature*

Land surface temperature was selected and taken into consideration as a parameter in this study as potential land suitability for cattle production was determined by the warm and cold climates of the study area, which were calculated from LST. The result revealed that high and medium temperatures were

more suitable than low temperatures. Hence, areas with 28°C to 31°C and greater than 31°C were highly suitable; however, less than 15°C was not suitable for cattle production in the study area. Spatially, the central and some northern parts of the study area were highly suitable than the southwestern part of the study area (Figure 3b). The result of this study is more consistent with (Akpoti et al., 2019; Sulieman, 2024).

### **Agro ecological zone**

The AEZ was the key determinant factor for land suitability for cattle production in some selected districts. The result shows that middle land (*woina dega*) and lowland (*kola*) agro climates were more suitable than highland (*dega*) agroclimates for cattle production in the study area. Most parts of the study area were highly suitable, except for some parts of the northern and southwestern parts of the study area (Figure 3c). Middle land and lowland were more suitable than highland areas in the Nile basin of Ethiopia.

### **Rainfall**

The availability of sufficient amounts of rainfall were another key factor in the identification of suitable land for cattle farming activities. The extent of existing rainfall causes the presence of various land suitability classes in the study area. Based on this reality, the areas with greater than 1582mm, 1455 to 1582mm, 1320 to 1455mm, and 1121 to 1320mm were classified as highly suitable, moderately suitable, marginally suitable, and not suitable sites for cattle farming activities in the study area (Ritzema et al., 2017). From the geographical distribution, a southern part of the study area was dominated by highly suitable areas, whereas an eastern part of the study area was occupied by unsuitable areas for cattle farming activities (Figure 3d).

## **Parameters used for land suitability for sheep production**

### **Slope**

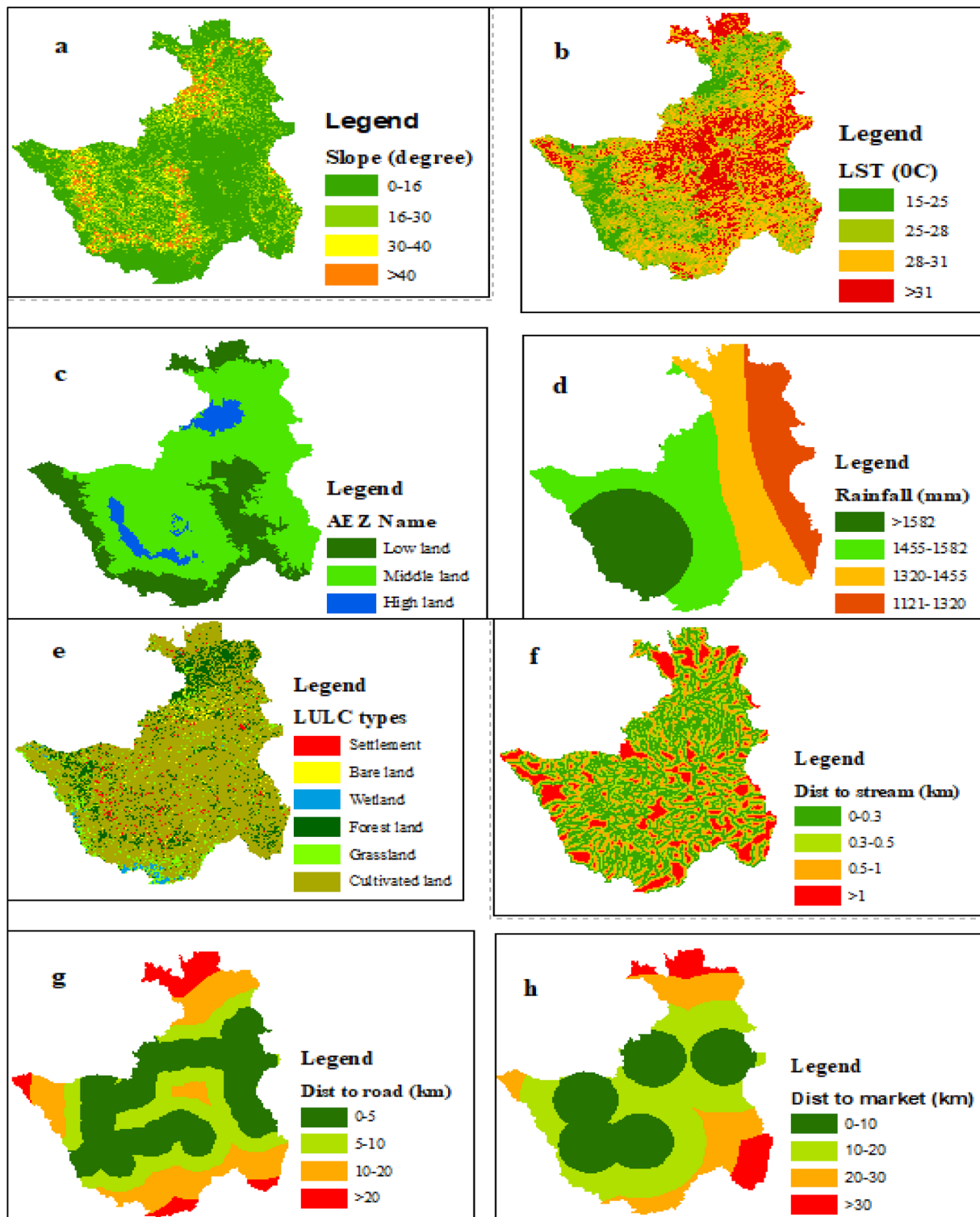
The steepness of the land feature was another key factor in the evaluation of the land's suitability for sheep farming in the study area. Hence, slope measured in degrees, land features with 0° to 16°, 16° to 30°, 30° to 40° and greater than 40° were ranked as highly suitable, moderately suitable, marginally suitable, and not suitable areas, respectively. Spatially, the south-eastern parts of the study area were dominated by highly suitable areas, whereas the central parts were occupied by moderately suitable areas for these activities. However, the south-western and northern parts of the study area were not suitable for sheep farming (Figure 4a). This result was consistent with the previous studies (Aggarwal et al., 2022).

### **Land surface temperature**

The LST of the study area was determined in degrees. The result reported that areas with 15°C to 25°C, 25°C to 28°C, 28°C to 31°C, and greater than 31°C were classified as highly suitable, moderately suitable, marginally suitable, and not suitable, respectively, for sheep farming activities. Some parts of the south-western and northern parts of the study area were occupied by highly suitable and moderately suitable areas, whereas some parts of the south-east were dominated by marginally suitable areas (Anachinaba et al., 2022). However, the central parts of the study area were not suitable for these farming activities (Figure 4b).

### **Agro ecological zone**

Agro-ecologically, the study area was classified into high land (*dega*), middle land (*weyina dega*), and low land (*kola*) areas. The result revealed that highland was highly suitable, whereas middle land was a moderately suitable area for sheep farming activities. However, low-land parts of the study were not suitable for sheep farming (Duguma & Duguma, 2022; Zhang et al., 2025). Geographically, the northern and south-western parts of the study area were highly suitable, whereas the south-eastern parts were not suitable for these farming types (Figure 4c).



**Figure 4.** Parameters used for sheep farming land suitability.

### *Rain fall*

The availability of rainfall was another key factor in identifying suitable land for sheep farming activities. Variation in the rainfall in the study area causes the presence of various land suitability classes for sheep farming activities. Based on this reality, the areas with greater than 1582mm, 1455 to 1582mm, 1320 to 1455mm, and 1121 to 1320mm were classified as highly suitable, moderately suitable, marginally

suitable, and not suitable sites for these activities in the study area (Anachinaba et al., 2022). From a geographical standpoint, the Southern part of the study area was dominated by highly suitable areas, whereas the eastern part of the study area was occupied by unsuitable areas for sheep farming activities (Figure 4d).

### ***Land uses and land cover types of the study area***

#### ***Cattle and sheep***

Sustainable cattle farming is achieved by selecting appropriate LULC types aligned with recommended suitability classes. The LULC types that exist in the study area were classified into different suitability classes like highly suitable ( $S_1$ ), moderately suitable ( $S_2$ ), marginally suitable ( $S_3$ ), and not suitable ( $N_1$ ), respectively. The result revealed that grass land and wetland are highly and moderately suitable, respectively, whereas cultivated land is a marginally suitable site for cattle and sheep farming. However, forest land, settlements, and bare land were not suitable areas for both sheep and cattle rearing. Geographically, some northeastern and southwestern parts of the study area were classified as highly suitable, while moderately suitable areas were located in the south-eastern parts of the study area (Duffy et al., 2022). A marginally suitable area for cattle farming was dominated in the eastern, central, and some western parts of the study area. The north and some eastern parts of the study area are also not suitable for cattle farming (Figures 3e, 4e).

#### ***Goats and camel***

A grassland LULC type was a highly suitable area for improving sustainable goat farming. The cultivated land area was marginally suitable, whereas settlements, bare land, forest land, and wetland were not suitable for goat and camel farming. The geographical distribution of highly suitable areas was suited to some south-eastern and some central parts of the study area, while marginally suitable areas were distributed across western and other central parts of the study area (Balew et al., 2022). Eastern and northern parts of the study area were not suitable for goat and camel farming due to the higher dominance of forest land and wetland LULC types (Figures 5e, 6e).

#### ***Proximity to stream***

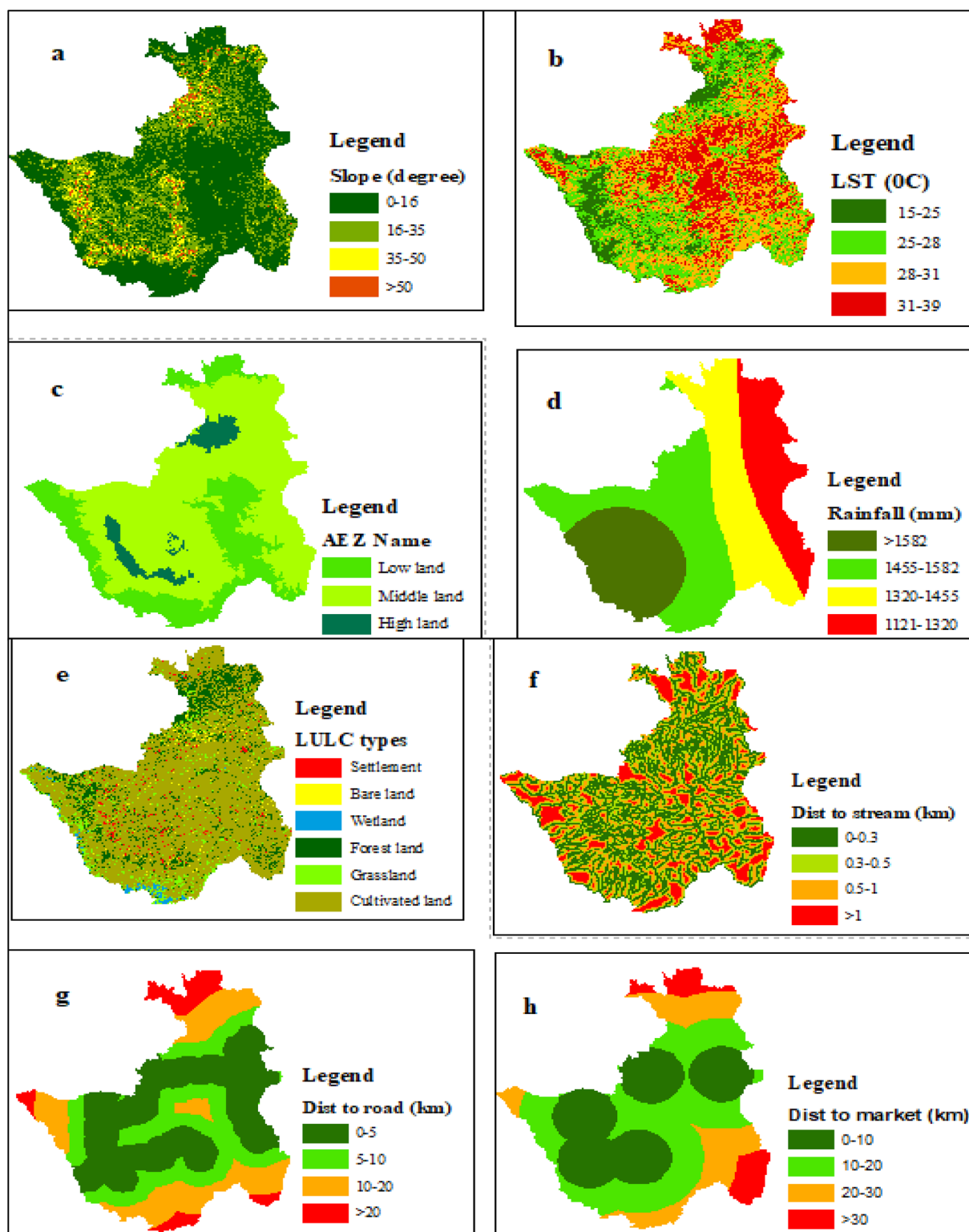
Availability and proximity to a water source were other key parameters in the selection of an appropriate site for livestock farming. The area nearest to the water source was highly prioritized over that of the distant one.

### ***Cattle***

The result shows that an area with a distance of 0 to 0.5km, 0.5 to 0.8km, 0.8 to 1.5km, and greater than 1.5 km from the stream was highly suitable, moderately suitable, marginally suitable, and not suitable, respectively, for cattle farming practices (Alturk et al., 2022). Some parts of the south-eastern and northern locations of the study area were dominated by highly suitable areas, whereas some parts of the central, southwestern, and north-eastern locations of the study area were dominated by moderately suitable LULC types. Marginally suitable land use types were distributed across some parts of the central and western sides of the study area, while the tip of the north, south, eastern, and western parts were occupied with unsuitable LULC types for cattle farming (Figure 3f).

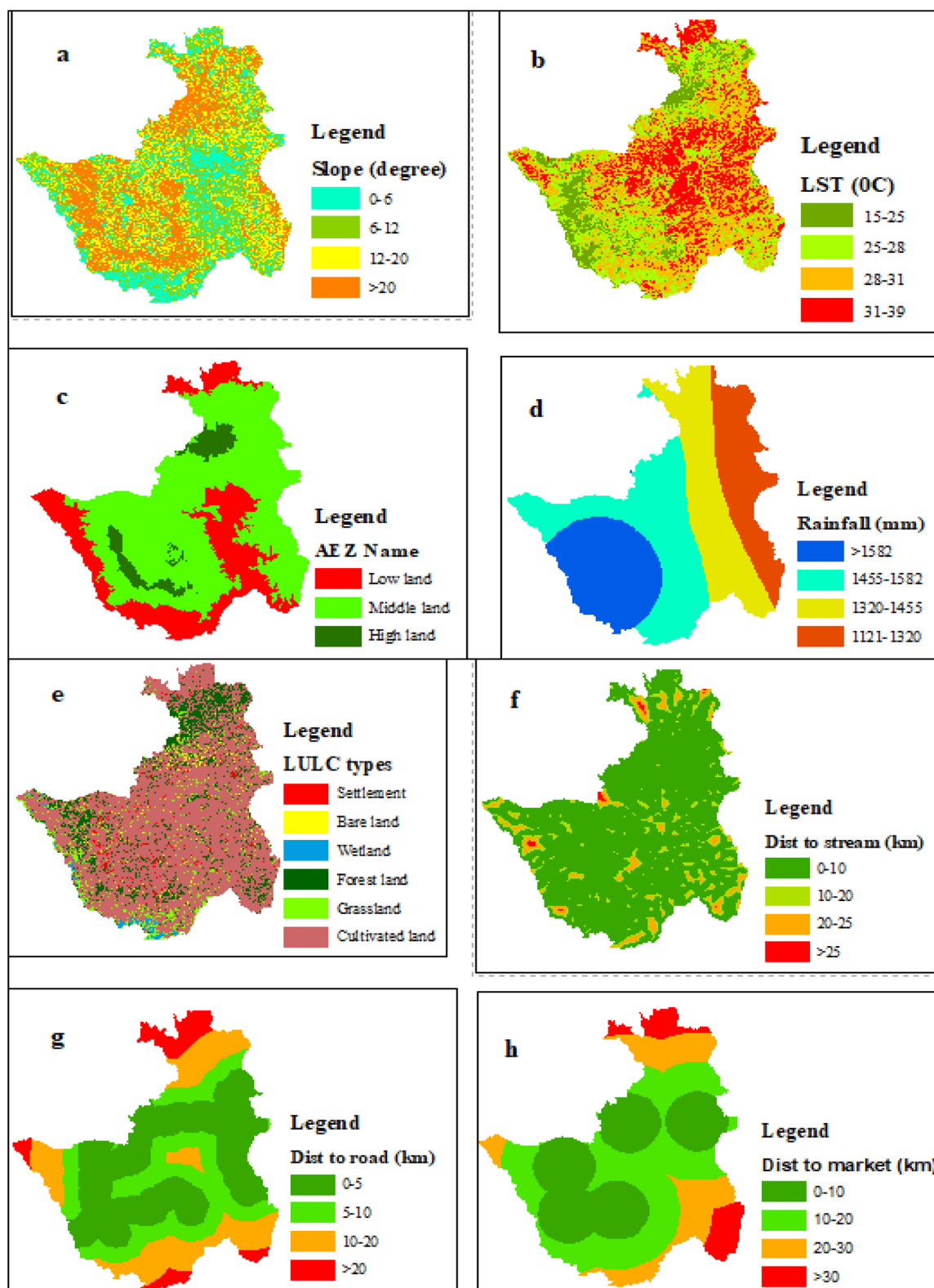
#### ***Goats and sheep***

Goats and sheep are small domestic ruminant animals that require an appropriate farming site in order to provide high-quality production. In the study area, there were various land suitability classes for goat and sheep farming based on distance to water sources. According to these suitability ranks, areas with 0 to 0.3km, 0.3 to 0.5 km, 0.5 to 1 km, and greater than 1 km distance to the water source



**Figure 5.** Parameters used for goat farming land suitability.

were classified as highly suitable, moderately suitable, marginally suitable, and not suitable consequently for goat and sheep farming activities (Nisan & Newas, 2022). Geographical locations of highly suitable areas were the northern and south-western parts of the study area, while some eastern and central parts of the study area were dominated by areas that were moderately suitable for this farming practice. This finding is consistent with the finding of Balew et al. (2022), who stated that most places



**Figure 6.** Parameters used for camel farming land suitability.

very close to perennial rivers are highly and moderately suitable for sheep and goat productivity. Consequently, some areas in the south and central parts were ranked as marginally suitable, whereas the south eastern, south western, and tips of the north parts were not suitable for goats and sheep out of the total study area (Figures 4f, 5f). This finding indicates that to increase the production and



productivity of goats and sheep, the pastoralists in the area should take into account the more suitable areas, thereby enhancing their livelihood incomes.

### **Camel**

There were opportunities for camel farming activities in the study area. However, area suitability may be influenced by distance to a water source. The result reported that areas with 0 to 10 km, 10 to 15 km, 15 to 25 km, and greater than 25 km distance from the stream were classified as highly suitable, moderately suitable, marginally suitable, and not suitable for camel farming sites (Rahimi et al., 2022). Spatially, some southern, western, and northern parts of the study area were dominated by highly suitable land, while some western and central parts were moderately suitable for camel farming. Consequently, marginally suitable and not-suitable sites were distributed to some northern, western, southern, and central parts of the study area (Figure 6f).

### **Commonly used parameters for all livestock farming**

#### **Proximity to road**

The sustainability of livestock production may be influenced by the inaccessibility of the road in the study area. The area nearest to the road gets more priority than the distant one as a suitable area for this production. Based on this reality, areas with a distance of 0 to 5 km, 5 to 10 km, 10 to 20 km, and greater than 20 km from the main road were ranked as highly suitable, moderately suitable, marginally suitable, and not suitable for livestock farming sites (Maru et al., 2022). Geographically, the western and eastern parts of the study area were highly suitable, whereas some central and southern parts were moderately suitable for livestock farming. Marginally suitable areas were distributed across the central and some southern parts of the study area, while the tip of the northern, southern, and western parts of the study were occupied by unsuitable areas for livestock farming (Figures 3g, 4g, 5g, and 6g).

#### **Proximity to market**

Livestock farming without market availability was not sustainable. Hence, the market was the central medium through which high-quality production obtained from livestock farming was sold at the right time and on the right coast. Aside from this, the distance from the farming site may cause variation in the suitability classes of the study area. The area with a distance of 0 to 10 km, 10 to 20 km, 20 to 30 km, and greater than 30 km from the market was classified as highly suitable, moderately suitable, marginally suitable, and not suitable for livestock farming (Benti et al., 2022). The spatial distribution of highly suitable areas was located in the northern and south-western parts of the study area, whereas some parts of the western and central parts were dominated by areas moderately suitable for livestock production. Some south-eastern and northern parts of the study area were occupied by highly suitable and not-so-suitable areas for livestock farming practice (Figures 3h, 4h, 5h, and 6h).

### **Parameters used for land suitability for goat production**

#### **Slope**

The suitability of land for goat farming in the study area was primarily determined by landscape conditions. Based on slope gradients, areas with slopes of 0° to 16°, 16° to 35°, 35° to 50°, and greater than 50° were classified as highly suitable, moderately suitable, and not suitable for goat-raising activities, respectively (Rahimi et al., 2022). Spatial analysis revealed that the central and some southern parts of the study area were highly suitable for goat farming, while the southeastern region was moderately suitable. Unsuitable areas were predominantly located in the western and northern parts of the study area (Figure 5a). The high suitability of central and southern areas can be attributed to gentler slopes, favorable vegetation cover, and adequate access to water resources, which are essential for goat grazing and health. Conversely, steeper slopes and less accessible terrain in the unsuitable zones limit forage availability and increase erosion risks, thereby reducing their viability for goat production. Understanding

these spatial patterns is crucial for targeted livestock management and optimizing resource allocation to enhance goat farming productivity sustainably.

### ***Land surface temperature***

The status of LST can influence the distribution of land suitable for goat farming activities in the study area. The result revealed that areas with LSTs of 15°C to 25°C, 25°C to 28°C, 28°C to 31°C, and 31°C to 39°C were classified as highly suitable, moderately suitable, marginally suitable, and not suitable for goat rearing activities (Unuvar & Dellal, 2022). The northern and western parts of the study were dominated by highly suitable and moderately suitable activities. However, the central parts of the study area were not suitable for goat farming (Figure 5b).

### ***Agro ecological zone***

Variation in the agro-ecological zone creates opportunities for the presence of different land suitability classes for goat farming. The result shows that lowland and middle land were highly suitable and moderately suitable, respectively, whereas highland areas were not suitable for goat farming activities in the study area (Mureithi et al., 2022). Geographically, the south-eastern and some central parts of the study area were dominated by highly suitable and moderately suitable areas, respectively, while the south-western and northern parts of the study area were not suitable for goat farming (Figure 5c).

### ***Rainfall***

Rainfall was another key environmental parameter for the evaluation of suitable land for goat farming activities within the study area. Variation in the rainfall in the study area causes the presence of various land suitability classes for these farming activities. Based on this reality, the areas with greater than 1582mm, 1455 to 1582mm, 1320 to 1455mm, and 1121 to 1320mm were classified as not suitable, marginally suitable, moderately suitable, and highly suitable sites for goat farming activities in the study area (Whannou et al., 2022). From the geographical distribution, the southern part of the study area was dominated by highly suitable areas, whereas the eastern part of the study area was occupied by unsuitable areas for goat farming activities (Figure 5d).

## ***Parameters used for land suitability for camel production***

### ***Slope***

Sustainable camel farming activities were slope-dependent activities in the study area. The result reported was that an area with a slope of 0° to 6°, 6° to 12°, 12° to 20° and greater than 20° was classified as highly suitable, moderately suitable, marginally suitable, and not suitable for camel farming practices (Alary et al., 2022). Geographically, the southern and eastern parts of the study area were highly suitable sites for camel farming. The western and southern parts were not suitable for these activities out of the total study area (Figure 6a).

### ***Land surface temperature***

Improvements in camel farming activities may influence LST in the study area. The result revealed that parts of the study area with LST of 15° to 25°, 25° to 28°, 28° to 31°, and 31° to 39° were ranked as highly suitable, moderately suitable, marginally suitable, and not suitable for this farming activity (Machan et al., 2022). Geographically, south western and some northern parts of the study area were highly suitable for camel farming. Excluding moderately suitable areas, the central parts of the study area were not suitable for sustainable farming activities (Figure 6b).

### ***Agro ecological zone***

In the three ecological zones of the study area, low-lying land was highly suitable for camel farming activities. However, the high land zone of the study area was not suitable for these activities due to the colder environmental conditions of the area (Bogale & Erena, 2022). Spatially, some of the central and

southern parts of the study area were highly suitable for camel farming activities. Unsuitable areas were distributed across some of the northern and south western parts of the study area (Figure 6c).

### Rainfall

Rainfall was another key environmental parameter for the evaluation of suitable land for camel farming activities in the study area. Variation in the rainfall in the study area causes the presence of various land suitability classes for these farming activities. Based on this reality, the areas with greater than 1582mm, 1455 to 1582mm, 1320 to 1455mm, and 1121 to 1320mm were classified as not suitable, marginally suitable, moderately suitable, and highly suitable areas for camel farming activities in the study area (Machan et al., 2022). From a geographical standpoint, the southern part of the study area was dominated by highly suitable areas, whereas the eastern part of the study area was occupied by unsuitable areas for camel farming activities (Figure 6d).

### Evaluation of potential land suitability for livestock production

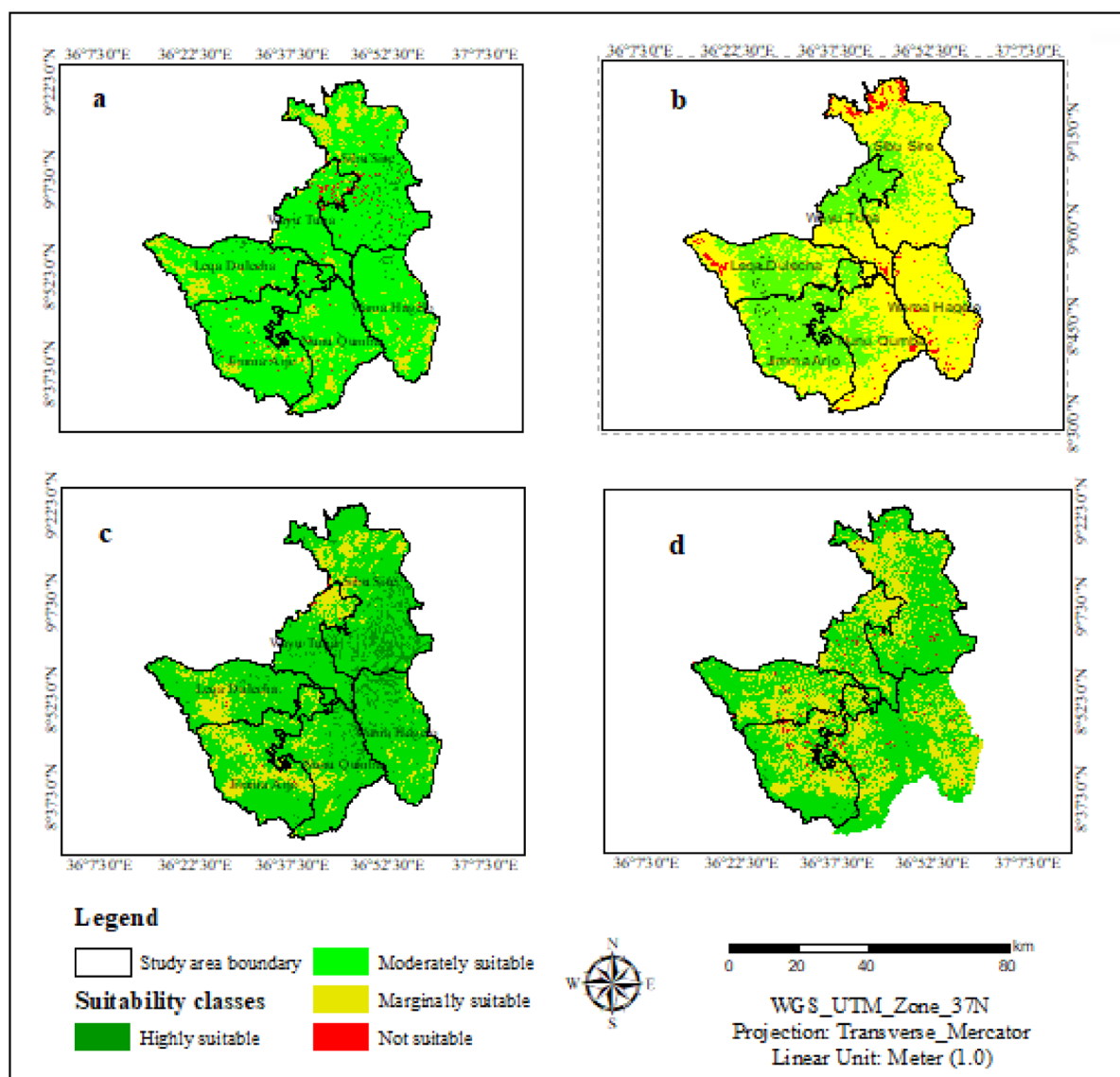
Eight determinant factors were aggregated for the evaluation of potential land suitability for livestock production. Those livestock were cattle, sheep, goats and camels.

### Land suitability analysis for cattle production

To evaluate potential land suitability for cattle production in the selected districts, the reclassified parameters were overlaid using ArcGIS environment. According to Getachew et al. (2022) studies, the suitability classes were categorized into highly suitable, moderately suitable, marginally suitable, and not suitable. The result revealed that, about 66.1 km<sup>2</sup> (1.6%) was highly suitable for cattle farming. Consequently, most parts of the study area was dominated by moderately suitable with area of 3504.3 km<sup>2</sup> (86.5%) for cattle production. However, about 48.6 km<sup>2</sup> (1.2%) was not suitable for cattle production out of the total study area due to limitation of determinant factors used in the study (Table 4). Moreover, the inclusion of factors such as distance to water sources, proximity to roads, and access to markets aligns with findings from other studies that emphasize the importance of infrastructure and accessibility in determining land suitability for livestock. Accordingly, Xie et al. in China noted that proximity to roads and markets played a crucial role in land suitability for livestock, a conclusion that resonates with the findings of the Ethiopian study, which incorporated similar logistical factors in its assessment. Spatially, in our study, the central and southern parts of Sibu Sire district, the northern parts of Wama hagalo and Leka dulecha districts, and the western parts of Wayu Xuqa district were dominated by suitable area for cattle farming. Consequently, the central parts of the selected districts for this study were occupied with moderately suitable land, while the western part of Sibu Sire and northern parts of Wayu Xuqa districts were not suitable for cattle farming (Figure 7a). Results AHP and GIS show that a significant portion of the land was classified as highly suitable for livestock farming. Although the percentage of highly suitable land in Bangladesh was higher than the 1.6% identified for cattle in Ethiopia, this discrepancy can likely be attributed to differences in local climate, terrain, and agricultural practices. The findings in Bangladesh underscore the role of regional factors in determining land suitability. Moreover, the result of this finding was more consistent with (Balew et al., 2022) which studied in Bale lowlands of Ethiopia and 4112 km<sup>2</sup> was occupied with highly suitable for cattle production.

**Table 4.** Suitable area for cattle farming in the study area.

| S/No | Suitability classes | Area (km <sup>2</sup> ) | Area (%) |
|------|---------------------|-------------------------|----------|
| 1    | Highly suitable     | 66.1                    | 1.6      |
| 2    | Moderately suitable | 3504.3                  | 86.5     |
| 3    | Marginally suitable | 430.2                   | 10.6     |
| 4    | Not suitable        | 48.6                    | 1.2      |
|      | Total               | 4049.2                  | 100.0    |



**Figure 7.** Suitable area for cattle (a), sheep (b), goat (c), and camel (d).

**Table 5.** Suitable area for sheep farming in the study area.

| S/No | Suitability classes | Area (km <sup>2</sup> ) | Area (%) |
|------|---------------------|-------------------------|----------|
| 1    | Highly suitable     | 25.8                    | 0.6      |
| 2    | Moderately suitable | 1449.3                  | 35.8     |
| 3    | Marginally suitable | 2482.2                  | 61.3     |
| 4    | Not suitable        | 92.0                    | 2.3      |
|      | Total               | 4049.2                  | 100.0    |

### Land suitability analysis for sheep production

Sheep production was crucial for solve food insecurity in the western parts of Ethiopia. According to the results, about 25.8km<sup>2</sup> (0.6%) was highly suitable and 1449.3km<sup>2</sup> (35.8%) was the moderately suitable for sheep farming in the study area. However, marginally was the most dominant suitability class due to the less conducive environment for sheep production in the selected districts. Consequently, about 92km<sup>2</sup> (2.3%) was not suitable for sheep production (Table 5). Geographically, the northern part of Wayu Xuqa, the southern parts of Leka dulecha and the western parts of Jimma arjo districts were highly suitable for sheep farming. In addition, most parts of Jimma arjo except some southern parts

**Table 6.** Suitable area for goat farming in the study area.

| S/No | Suitability classes | Area (km <sup>2</sup> ) | Area (%) |
|------|---------------------|-------------------------|----------|
| 1    | Highly suitable     | 315.3                   | 7.8      |
| 2    | Moderately suitable | 3115.3                  | 76.9     |
| 3    | Marginally suitable | 612.3                   | 15.1     |
| 4    | Not suitable        | 6.4                     | 0.2      |
|      | Total               | 4049.2                  | 100.0    |

**Table 7.** Suitable area for camel farming in the study area.

| S/No | Suitability classes | Area (km <sup>2</sup> ) | Area (%) |
|------|---------------------|-------------------------|----------|
| 1    | Highly suitable     | 20.1                    | 0.5      |
| 2    | Moderately suitable | 2603.8                  | 64.3     |
| 3    | Marginally suitable | 1360.3                  | 33.6     |
| 4    | Not suitable        | 65.0                    | 1.6      |
|      | Total               | 4049.2                  | 100      |

of Jimma Arjo, the central parts of Leka dulecha, the western parts of Nunu Qumba, the northern parts of Wayu Xuqa and the western parts of Sibu Sire districts were moderately suitable. However, northern parts of Sibu sire, western parts of Leka dulecha, the western parts of Wama Hagalo and eastern parts of Nunu Qumba, boundaries between all selected districts were not suitable for sheep production (Figure 7b). Variation in suitability classes across different regions is the cumulative effects of local environmental conditions and management practices influencing the outcomes of land suitability assessments. The result of present study was in line with the previous of Karthik et al. (2021), cold temperatures and high land area were highly suitable, while a warm climate and low land area was not suitable for sheep production.

### Land suitability analysis for goat production

From the reclassified overlaid parameters by using ArcGIS environments, the targeted districts final potential suitability areas for goat farming were identified. Those identified suitable area classified as highly suitable, moderately suitable, marginally suitable and not suitable classes based on the final output obtained from overlaid determinant factors. The results of this study show that about 315.3 km<sup>2</sup> (7.8%) area was highly suitable for goats farming whereas 3115.3 km<sup>2</sup> (76.9%) area was dominated with moderately suitable for goat farming activities. Consequently, about 612.3 km<sup>2</sup> (15.1%) area was occupied with marginally suitable while about 6.4 km<sup>2</sup> (0.2%) area was not suitable out of the total study area (Table 6). From the geographical information's, the southern parts of Sibu Sire district, the northern parts of Nunu Qumba and northern parts of Leqa Dulecha district were occupied by highly suitable area for goats farming activities. Beside this, most central parts of the study area excluding marginally suitable area were dominated with moderately suitable for goats farming. However, the central parts of Jimma Arjo and northern parts of Wayu Xuqa were not suitable for this production due to high rainfall, low temperature and high elevated area (Figure 7c). The final output of this finding was more consistent with the previous studies (Habibie et al., 2021).

### Land suitability analysis for camel production

Camel is one of the livestock that provides food security in Ethiopia and is highly produced in the warm climate and lowland areas. The aggregated factors, the final land suitability for camel production was produced. The result shows that, 20.1 km<sup>2</sup> (0.5%) was highly suitable and 2603.8 km<sup>2</sup> (64.3%) was moderately suitable for camel production in the selected districts. In addition, about 1360.3 km<sup>2</sup> (33.6%) and 65 km<sup>2</sup> (1.6%) were marginally suitable and not suitable, respectively (Table 7). Spatially, the western parts of Jimma Arjo district, particularly Arjo Dhidhessa rift valley, were highly suitable for camel production. The central parts of the study were marginally suitable or not suitable for camel production (Figure 7d). According to the results, lowland (*kola*) agro climate, low rainfall and high temperatures were highly suitable, whereas highland (*dega*) agro climate, high rainfall and low temperature were not suitable for

camel production. Al-Abdulla (2021) stated that, warm climate and lowland area were highly suitable for camel production than other. Generally, this study adds to the expanding body of evidence supporting the integration of AHP and GIS as effective tools for land suitability assessment. The results for Ethiopia, especially concerning the land use potential for various livestock species, align with findings from comparable studies worldwide. These outcomes indicate that, through the implementation of targeted land management practices and policy measures, the identified suitable areas can be effectively utilized to enhance livestock production in the region, mirroring successful strategies employed in other parts of the world.

## Conclusions

Livestock production was crucial for developing income for society and solving food insecurity in Ethiopia. In this study, geospatial technologies with MCE were applied to analyze potential land suitability for livestock production. Based on the identified problem, six districts namely Sibu Sire, Wayu Xuqa, Leka Dulecha, Jimma Arjo, Nunu Qumba, and Wama Hagalo were selected to evaluate land suitability for cattle, sheep, goat, and camel production. For this study, eight parameters like slope, LST, agro-ecological zone, rainfall, LULC, distance from streams, distance from roads, and distance from markets were used to determine potential land suitability for livestock production. All factors were reclassified according to their suitability for cattle, sheep, goats, and camel farming. To determine the influence of the parameters, a pair-wise comparison was applied. The result shows that 1.6%, 0.6%, 7.8%, and 0.5% were highly suitable for cattle, sheep, goat, and camel production, respectively. Consequently, gentle and steep slopes, middle land (*woina dega*) and highland (*dega*), medium and low temperatures, high rainfall, and medium rainfall were highly suitable for cattle and sheep production. This shows that similar environments were suitable for cattle and sheep farming. However, the gentle slope, high temperature, low rainfall, and lowland (*kola*) agro climate were highly suitable for goat and camel production. In addition, short distances from a stream, road, or market and area coverage by grassland were highly suitable for cattle, sheep, goat, and camel production, and vice versa. Based on the study findings, it is recommended that farmers, livestock specialists, researchers, and policymakers leverage the identified suitable areas to optimize sustainable livestock production within the selected districts. Future research should integrate socio-economic variables such as market dynamics and livestock management practices alongside climate change projections to enhance the robustness and predictive accuracy of land suitability models.

## Author's contributions

MBM and MMG participated in research design, document analysis and manuscript writing. DGO and MLD participated in data collection, methodology, data analysis and interpretation. KTD, ZRR, TT and DOG participated in research design, literature review, data analysis and final draft edition. All authors read and approved the final manuscript for publication.

## Consent for publication

The authors agreed to publish this manuscript on Journal of Cogent Food and Agriculture.

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## Disclosure statement

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## Data availability statement

Available in the manuscript.

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